

Case Study-Excel-Power Query

Customer Churn Analysis

An end-to-end analysis of 2,800 subscription customers — identifying why customers leave and what drives retention across Basic, Standard, and Premium plans.

2800

Total Customer

2023-2024

Time Period

Kaggle

Data Source

Excel-Power Query

Tools Used

Project Overview

In this project, I created a customer churn dashboard to analyze customer subscription behavior, usage patterns, and churn trends. The dashboard helps businesses understand why customers leave their services and how customer engagement affects retention.

About Data

I imported the dataset from Kaggle, which contained 10 columns and 2,800 rows. After importing the data, I cleaned the messy dataset using Power Query to improve data quality and consistency. To perform time-based analysis, I extracted the Month and Year from the dataset, which helped me analyze customer churn and behavior trends over different periods.

Problem Statement

Companies often struggle to understand why customers stop using their services. Although the data contains valuable insights, it is difficult to identify customer churn, engagement trends, revenue performance, and payment or support issues without proper analysis. The challenge was to create a dashboard that simplifies complex data and provides clear, actionable business insights.

Project Goal

The objective of this project was to analyze customer churn patterns, monitor customer usage behavior, compare subscription plan performance, identify factors affecting customer retention, and create an interactive dashboard that provides clear business insights

Customer Churn Dashboard

Plan

Basic

Premium

Standard



Total Customer

2800

2023-2024



Churned Customer

1605

57.32% Churned Rate



Total Revenue

₹ 12,15,800

All plan combined



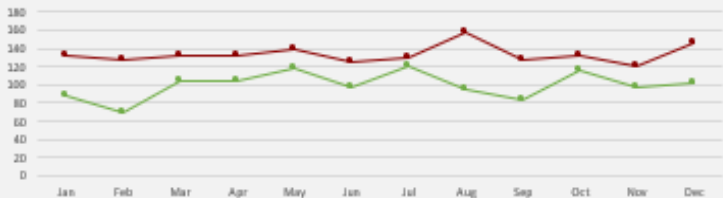
Revenue Lost

₹ 6,97,995

57.41% of Total Revenue

Monthly Churn Trend

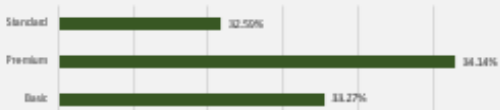
Churned Retained



Avg. weekly usage hours

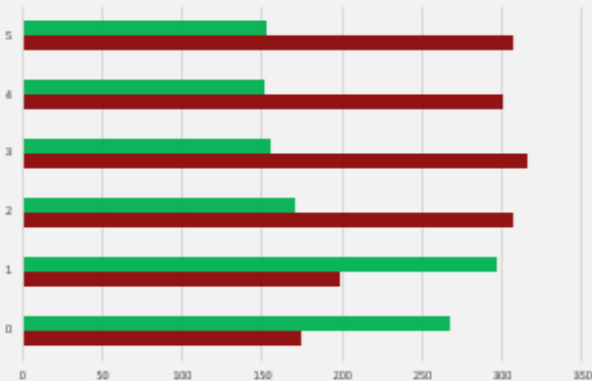


Churn rate by plan



Churn vs payment failures

Retained Churned



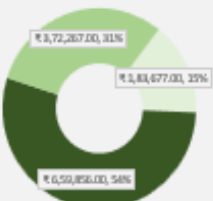
Churn by support tickets raised

Churned Retained



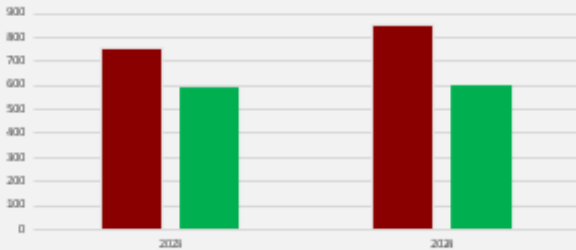
Revenue Distribution

Basic Premium Standard

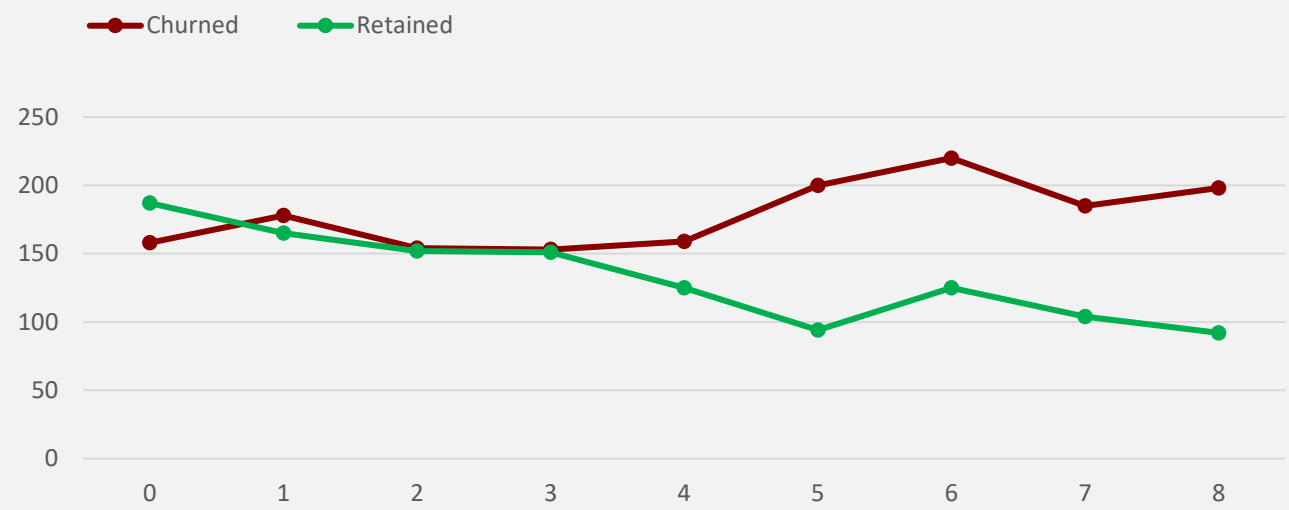


Churn By Year

Churned Retained



Churn by support tickets raised



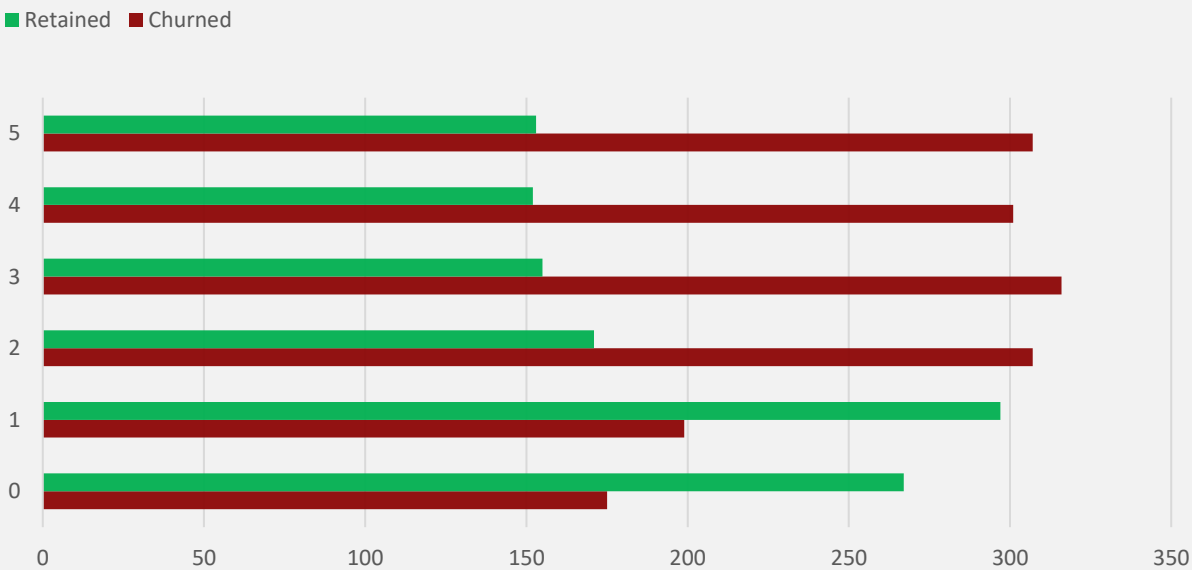
More Support Ticket = More Churn

Customers who raised 0 support tickets show the highest retention, while those raising 7–8 tickets show the highest churn counts. High ticket volume signals unresolved pain customers try to get help, then leave.

More Payment Failures, Higher Churn Risk

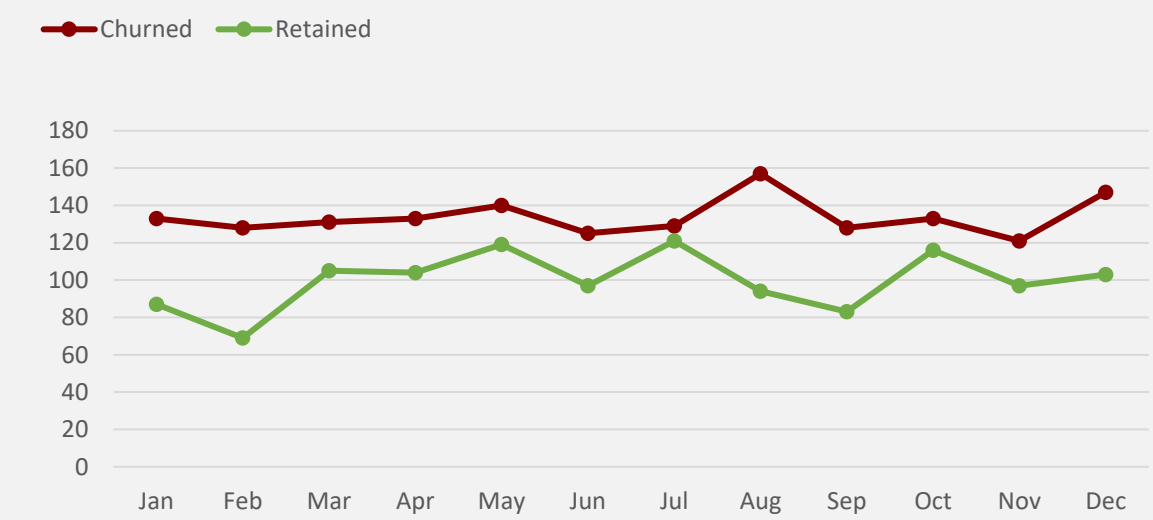
Customers with more than 2 payment failures are more likely to leave the service. As payment failures increase, customer churn also rises significantly. This shows that repeated payment issues are a strong sign of customer loss.

Churn vs payment failures



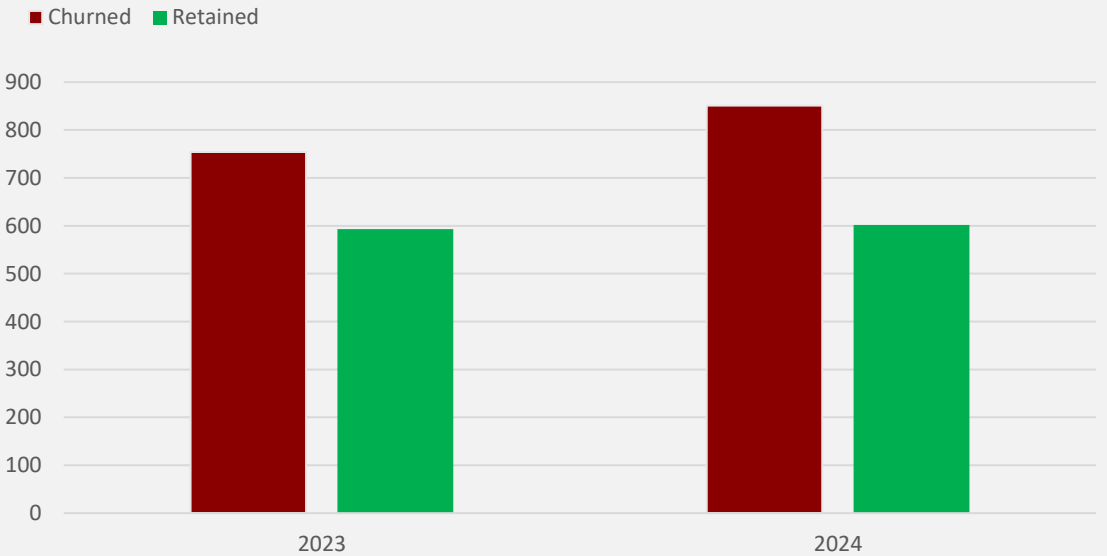
August is the highest-churn month (157 churned)

Monthly Churn Trend

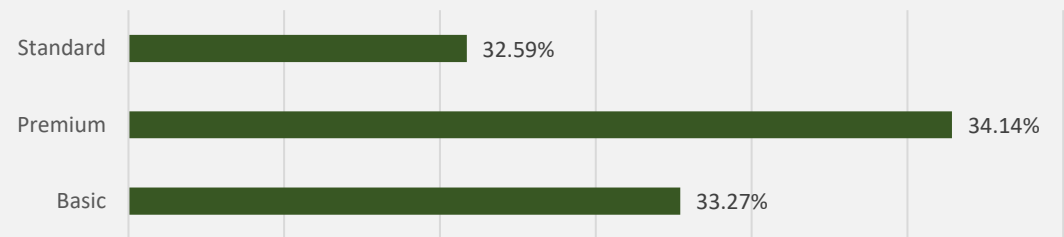


In 2024 shows a 12% rise in churned customers

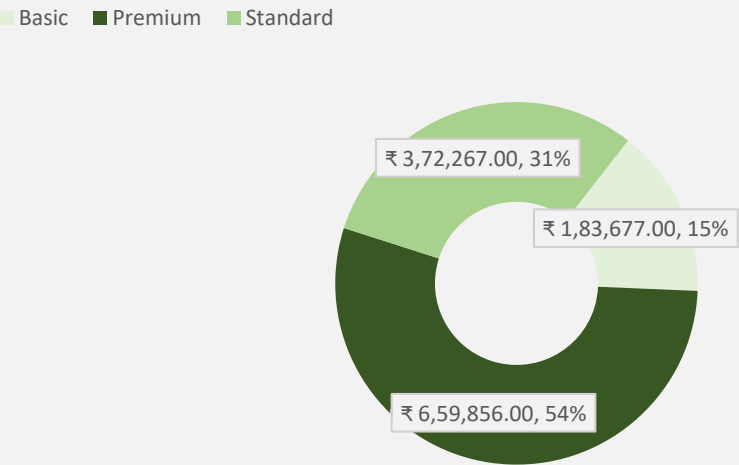
Churn By Year



Churn rate by plan



Revenue Distribution



Premium Plan Shows Strong Revenue but Poor Retention

Premium plan generates more revenue than the Basic and Standard plans, but it also has the highest churn rate. Despite paying more, Premium users show lower engagement and are more likely to leave the service. This suggests the plan may not be delivering enough value compared to its cost.

₹

Total Revenue

₹ 12,15,800

All plan combined



Revenue Lost

₹ 6,97,995

57.41% of Total Revenue

Key finding: Over half of all revenue (₹6.98L of ₹12.16L) was lost to churned customers — making churn the single biggest threat to business sustainability.

Recommendation

- Improve engagement for Premium plan customers through better communication and useful features
- Improve customer support by solving issues faster and reducing repeated complaints.
- Reduce payment failures by sending reminders and adding easy payment options
- Check the reason for the high churn in August and plan retention activities during that period
- Identify customers at risk early based on payment failures, support tickets, and low usage to reduce churn.

Conclusion

I analyzed customer behavior by comparing churn status with payment failures, support tickets, and subscription plans. Through visual analysis, I found that repeated payment failures and higher support issues were major factors influencing customer churn. I also discovered that although the Premium plan generated higher revenue and customers, it showed higher churn, highlighting the need for better customer retention strategies.